AIVANCIN

Your next role starts with the work you already do.

Describe what you actually do. AI suggests the matching work activities, you confirm them, and AIVANCIN shows the closest graduate roles and a free first plan.

Team DOVUN · German-Malaysian Institute
SDC Hackathon 2026 · Domain One: The Aiversecity Future-Ready Workforce

The problem

40.7%

of employed Malaysian graduates aged 25 to 34 work in semi-skilled or low-skilled jobs.

751,100 people

Before spending evenings and savings on retraining, they need to know: **which graduate job is closest to the work I already do?**

Source: DOSM, Quarterly Skills-Related Underemployment by Age, Q2 2026 (CC BY 4.0)

Meet Hafiz (a synthetic persona)

27 · Computing diploma · Delivery van driver

His job title says

Vehicle-operator roles

- Light Truck Drivers
- Supervisors of vehicle operators, crane operators

0 of the top 10 roles are in computing

His weekends say

- Fixes friends' laptops and home wifi
- Installs software, cleans out viruses
- Backs up phones and laptops
- Keeps his uncle's shop website updated

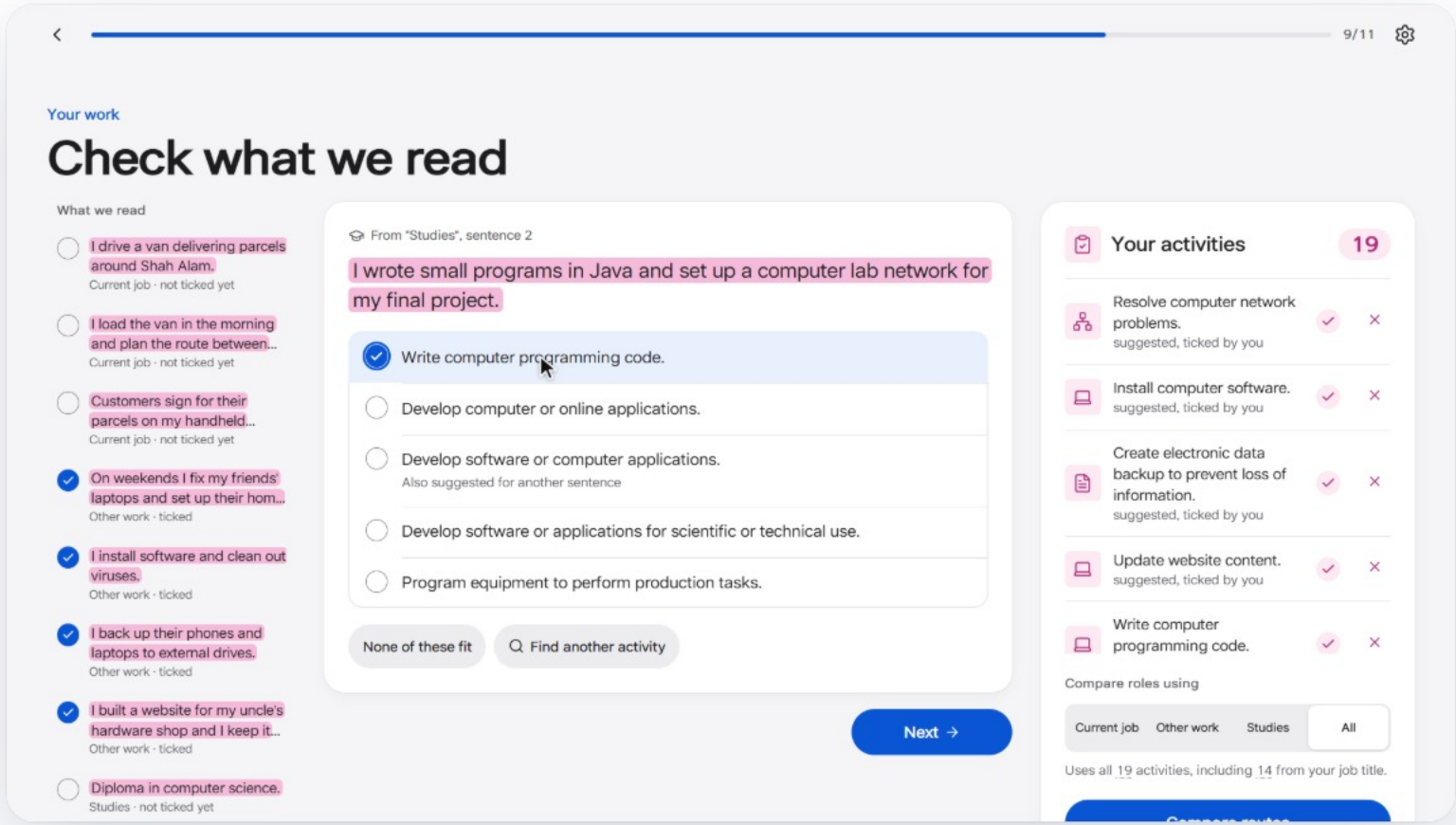
With that work confirmed: at least 6 of the top 10

His title hides this experience. Confirm it, then compare destinations.

Live prototype · Describe and confirm

Confirm what is true

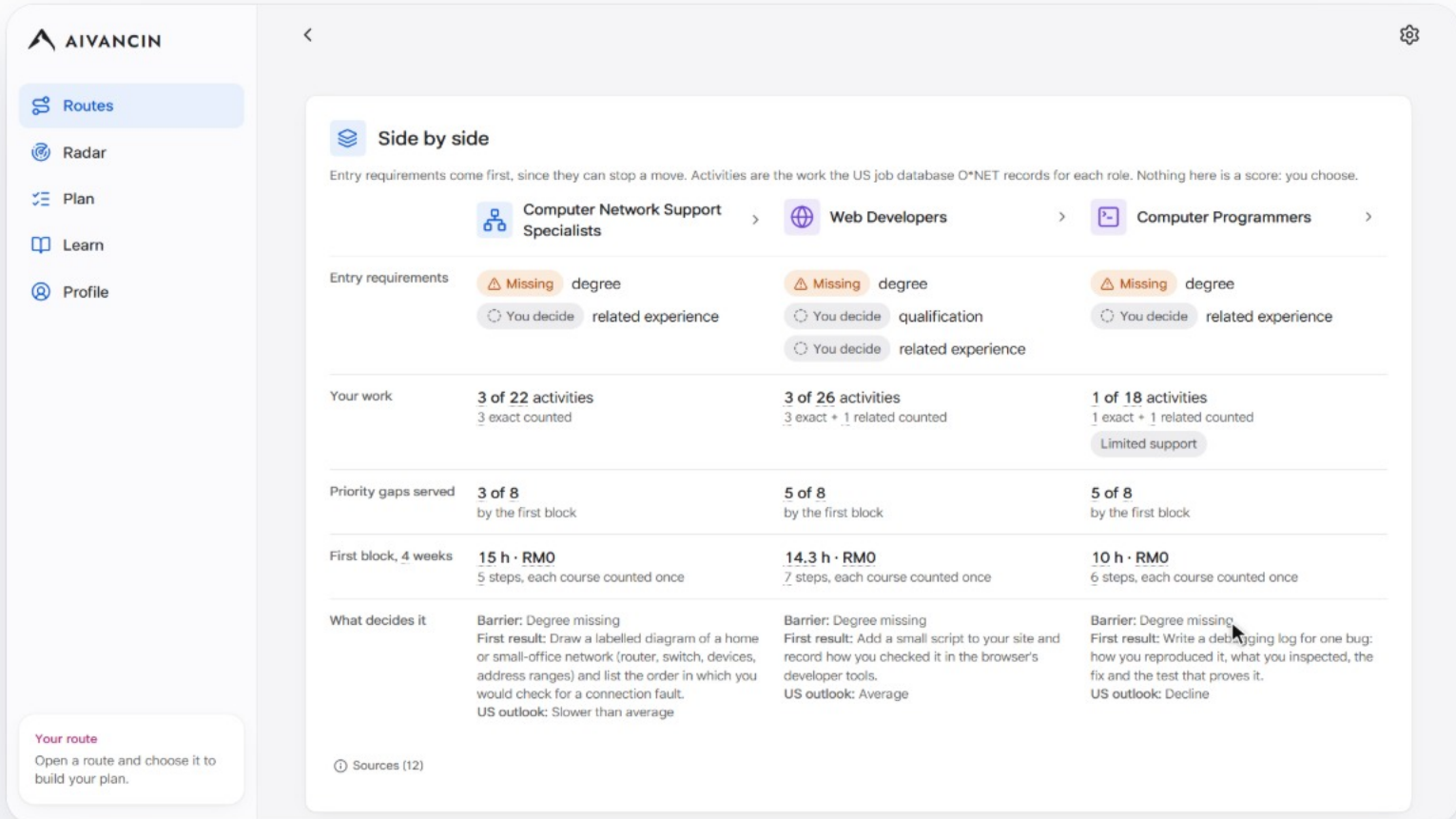
The encoder links descriptions to O*NET's 2,087 activities, with sentence evidence. Short sentences get five suggestions; long ones can get ten. You confirm what is true.



Live prototype · Compare

Routes that use your work

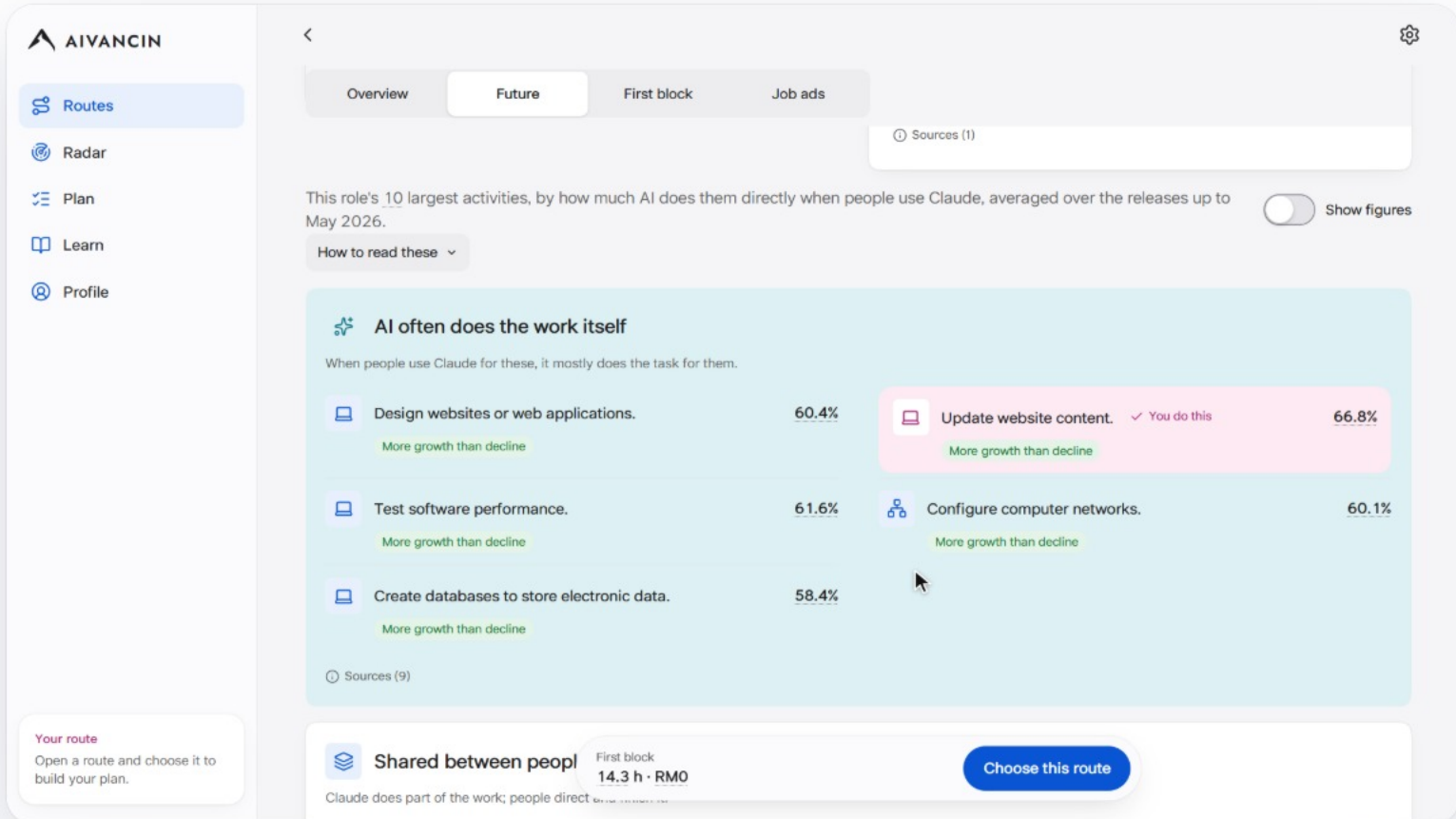
Roles ranked by activity overlap, with the work behind each route, the requirements still open and the gaps the first block covers. Reorder by outlook, entry or decent work.



Live prototype · Route detail

The future of each route

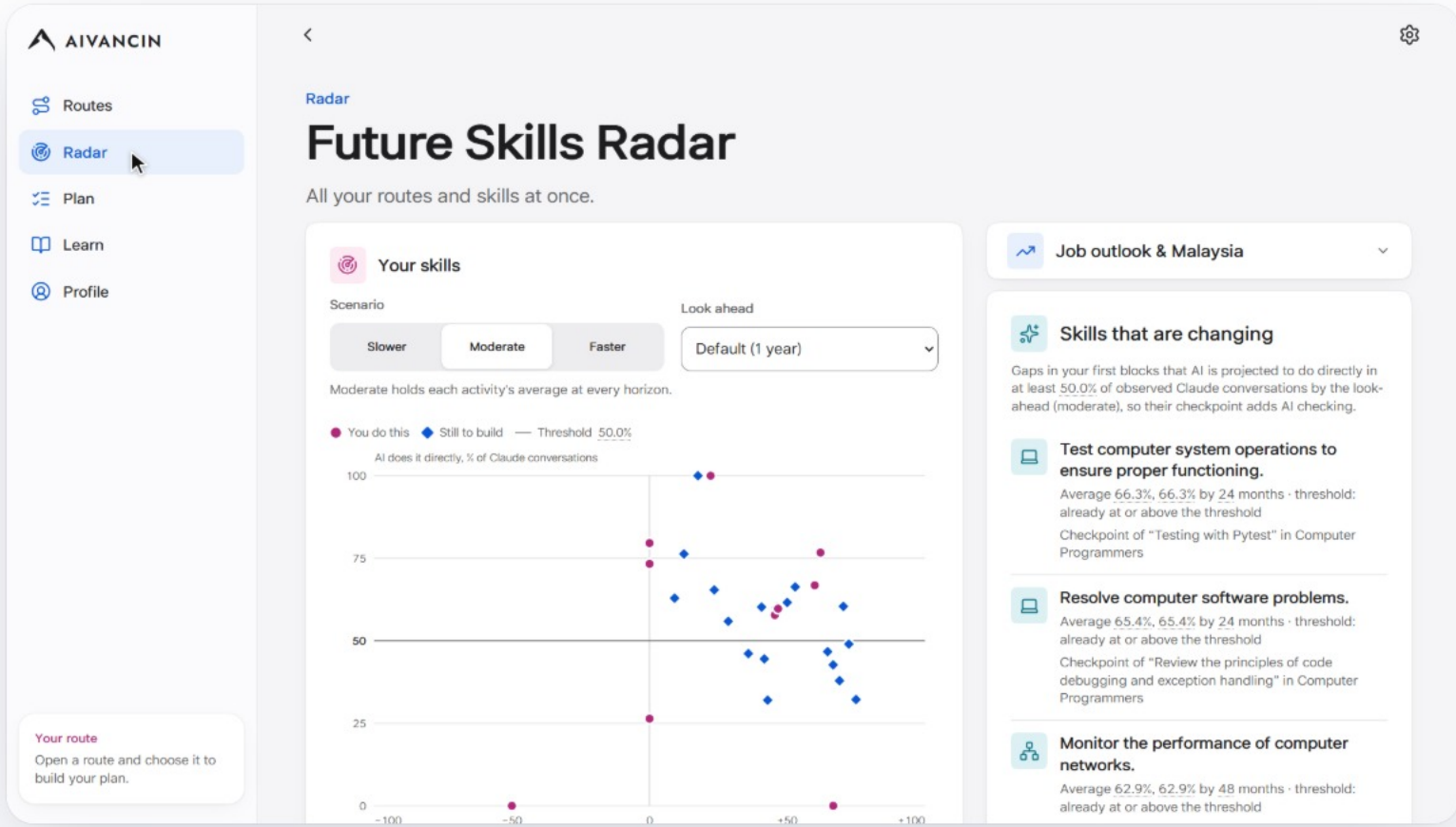
US projections from BLS with their accuracy record, Malaysia's observed DOSM trend, and the average and latest global Claude usage for each activity, under three scenarios.



Live prototype · Radar

Future Skills Radar

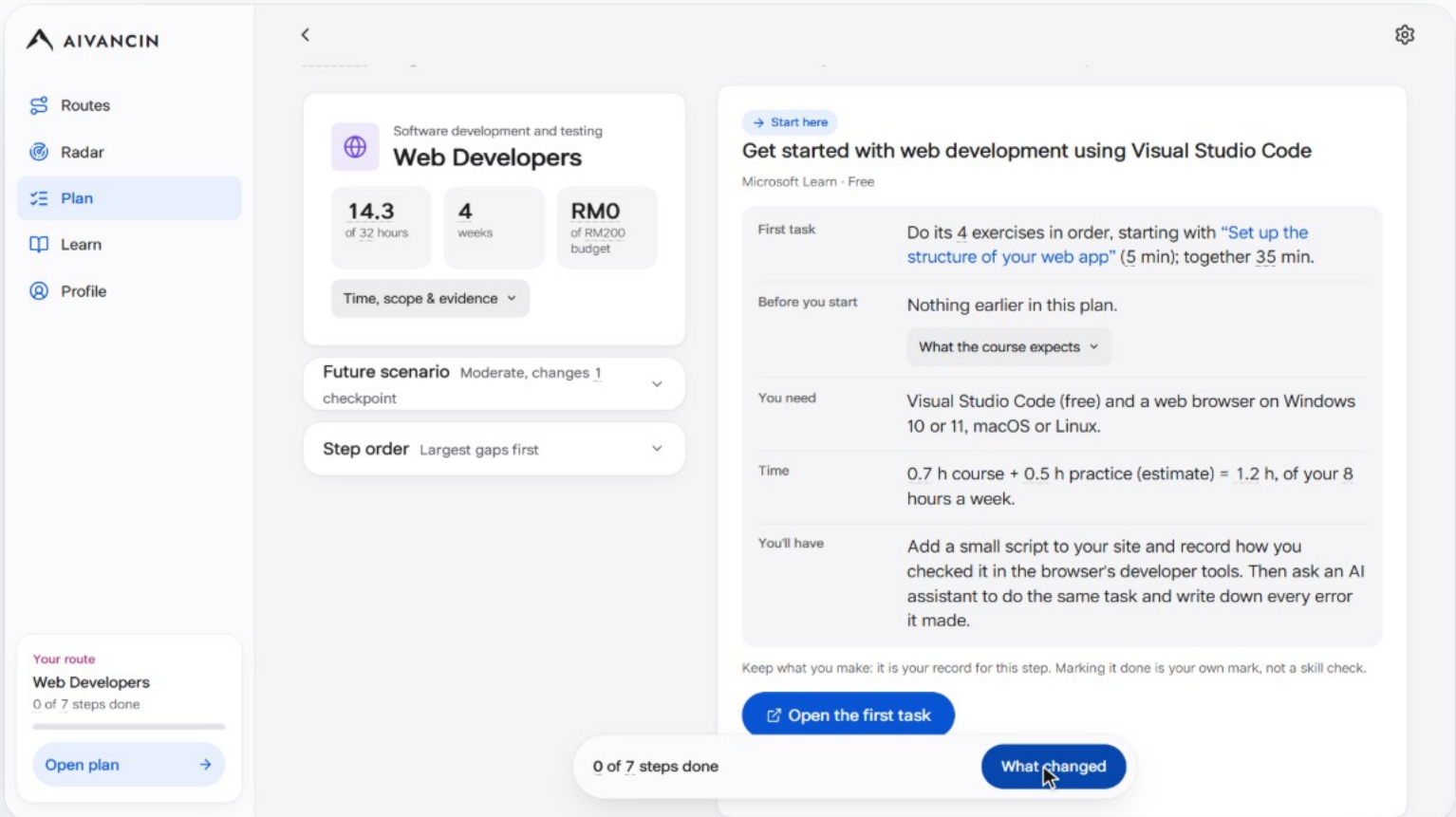
Your skills in pink, your gaps in blue. Across: growing or declining US demand. Up: the average direct-use share in Claude conversations. Scenarios are explicit; estimates carry their backtested error.



Live prototype · Plan

A free first learning block

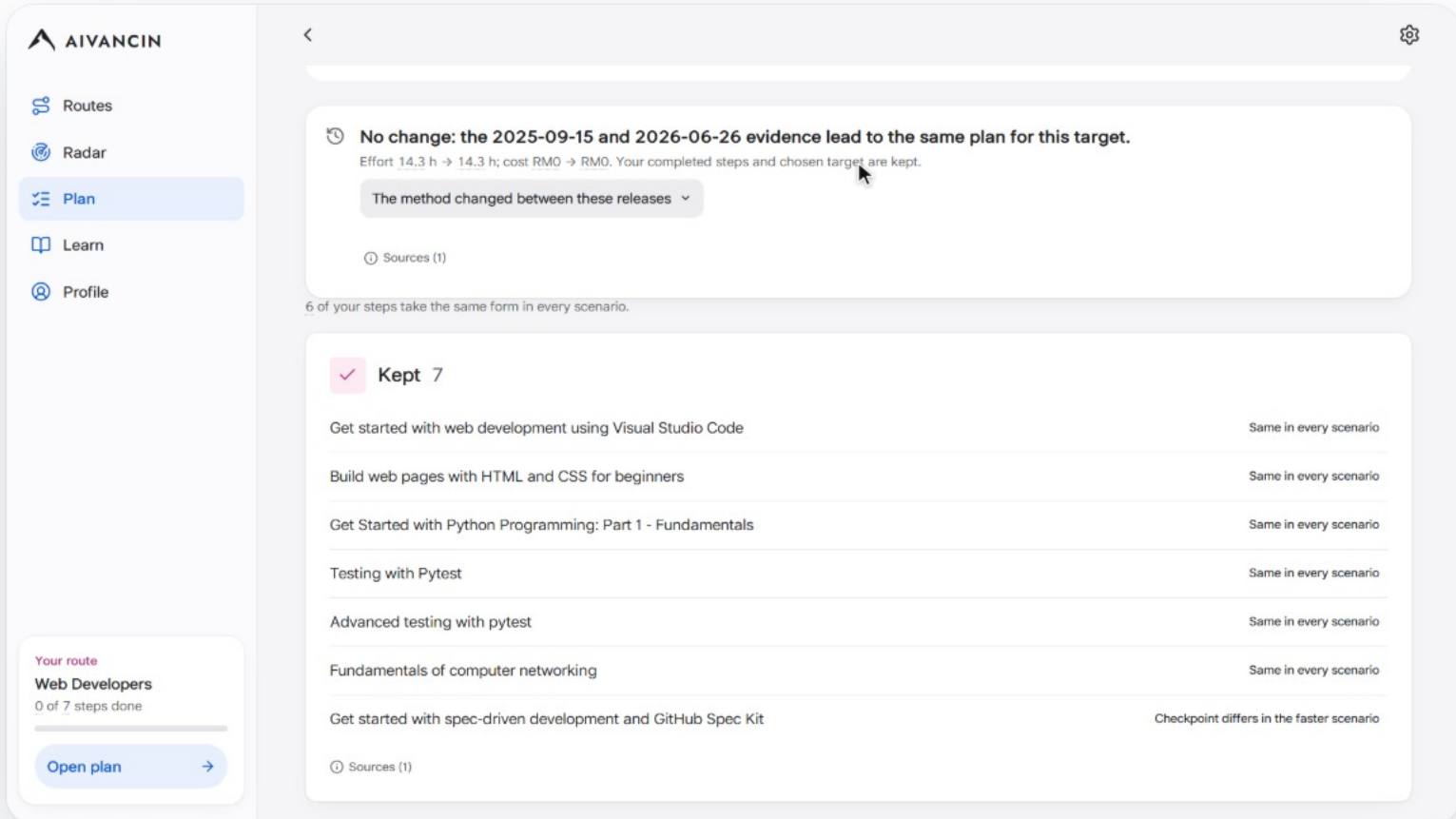
Start here names the exact unit, setup and output. The free four-week block respects time, prerequisites and budget; introductory and paid provider-exercise limits stay visible.



Live prototype · What changed

New evidence, your decision

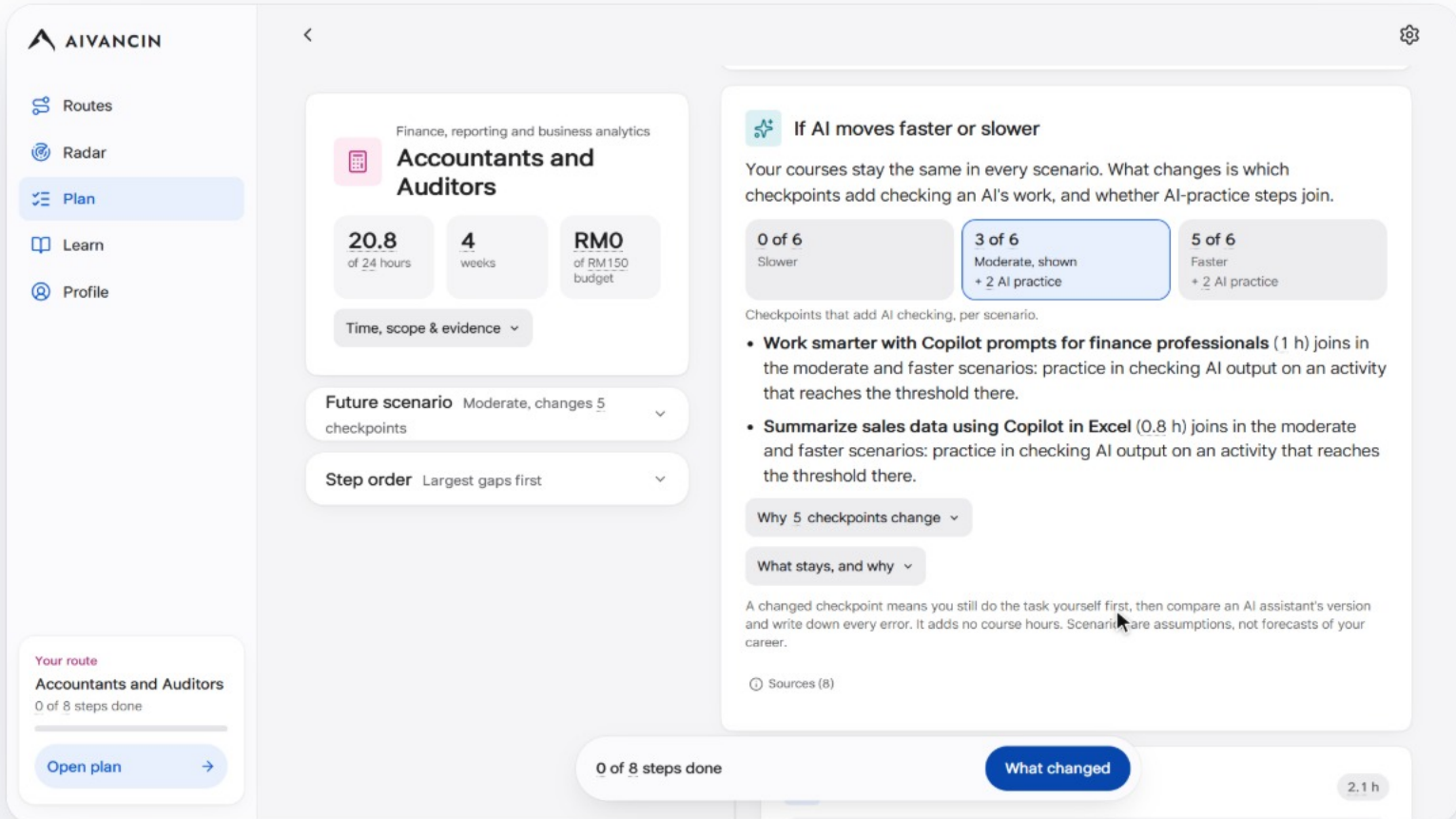
With Hafiz's work, target and progress fixed, the bundled releases lead to the same moderate-scenario plan. The app reports no change; it never forces one.



Prototype · Alternative assumptions

What changes in the plan

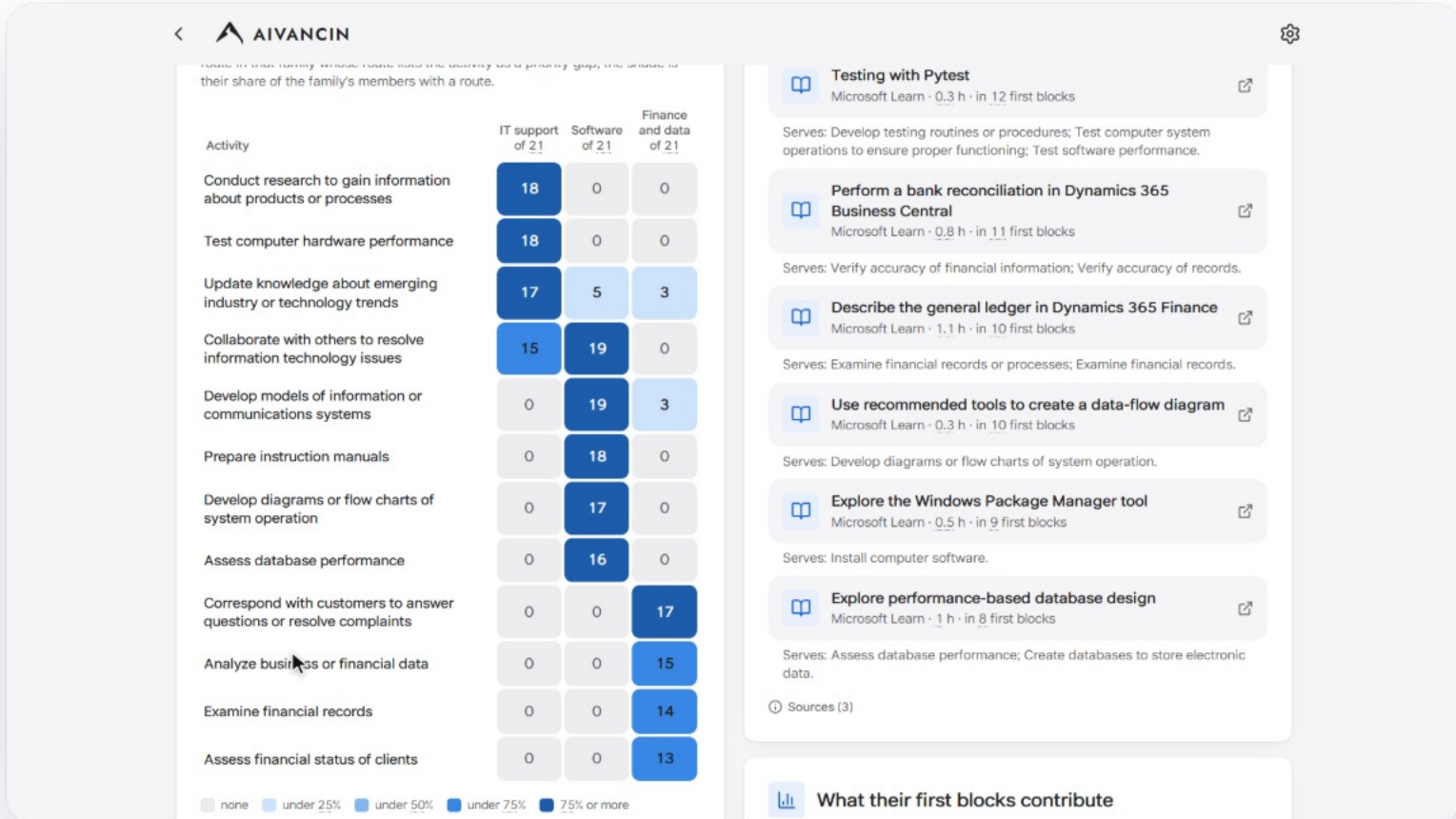
Priya, Accountants and Auditors: 0, 3 and 5 of 6 checkpoints add AI checking in the slower, moderate and faster scenarios. Two AI-practice steps join only in the latter two; foundations stay first.



Live prototype · Career centres

A cohort view

Shared priority gaps across a synthetic cohort of 60 graduates, by route family. Counts only, never a score; it changes no one's routes.



Domain One must-haves

All six run in the prototype.

1

Skills profile intake

Four questions in your own words. AI suggests activities; you confirm each one.

2

AI-trend signal layer

Four Anthropic Economic Index releases, BLS outlook and DOSM, each sourced and dated, with a source watch.

3

Future-role matching

Ranked by activity overlap, with requirements, gaps and the work behind each route.

4

Reskilling pathway

An ordered first block of verified free courses, sized to your hours and budget.

5

SDG mapping

Reasoned tags on every step (8.5, 4.4, 4.3, 8.2) and how your path contributes.

6

Handling uncertainty

Slower, moderate and faster scenarios; estimates with their backtested error.

How it works

A small model, checked at every step.

1. **You describe your work**
   plain sentences, your own words
2. **Fine-tuned MiniLM encoder**
   quantized ONNX on the CPU; 2,087 recorded activities
3. **You confirm what is true**
   nothing is accepted silently
4. **Deterministic matching and plans**
   no generated text; same input, same result
5. **Results with sources and dates**
   O*NET, Anthropic Economic Index, BLS, DOSM

RunPod trained

Fine-tuned once on an **RTX 4090** with O*NET task-to-activity pairs. Shipped because it passed a gate fixed before training: **45 of 60** held-out descriptions, against 39. Cost US$0.42.

Replit live

The prototype runs at **aivancin--jadfahmi.replit.app** from bundled, dated, hashed public data. No API keys, no per-request cost.

Evidence

What we measured, including what fails.

32 : 20

relevant roles from the confirmed model chain versus keyword search

Relevant top-five roles out of 50, identical synthetic text

Pass

45/60

held-out descriptions with a correct activity after fine-tuning

39 before; gate fixed before training

Shipped

7/12

in-scope profiles with two complete routes and blocks

58.3%, against our 80% target; keyword 2/12, title-only 0/12

Not yet met

Hardest risk: missed work and off-family routes, including within the supported fields. **Fallback:** honest exploration states and "no verified resource", never an invented job or course.

Predeclared synthetic set 8; first run and final rerun 7/12. No real-user outcome validation. 354 Python tests and 14 Node tests pass (23 Oct). Details in the README.

SDG impact

From the work you do to decent work.

Targets	**SDG 8.5** **SDG 4.4** Decent work for young people; relevant skills for employment. Supporting: 4.3 free learning, 8.2 working with AI.
Baseline	**40.7%** of employed graduates aged 25 to 34 in semi- or low-skilled jobs (751,100; DOSM, Q2 2026)
KPI	Route-and-block coverage: in-scope profiles with two skilled routes, sourced requirements and a free first block. **Newest set: 7 of 12 (58.3%)**
Phase 2 target	**At least 80%** on a fresh, predeclared profile set; then testing with real underemployed graduates

AIVANCIN

You choose.
AI suggests.

Live prototype aivancin--jadfahmi.replit.app

Code github.com/AIVANCIN/aivancin

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